

Lecture 22

Recap.

① Continuous R.V.

X is said to be continuous if

$$F_X(x) = \int_{-\infty}^x f_X(u) du$$

for some function $f_X(\cdot) \rightarrow$ pdf
probability density function

② Expectation of
a continuous R.V.

• Expectation of
function of a R.V.

③ Example R.V.s.

- uniform
- exponential
- normal R.V.

Properties of pdf

① $f_X(x) \geq 0 \quad x \in \mathbb{R}$

② $\int_{-\infty}^{\infty} f_X(x) dx = 1$

③ $P(X \in B) = \int_{x \in B} f_X(x) dx$

$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$

For non-negative R.V. X .

$$E[X] = \int_0^{\infty} P(X > x) dx \quad \Bigg\} \xRightarrow{\text{to prove}} E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

a) $X \sim \text{Uniform}[a, b]$

$$f_X(x) = \begin{cases} c & x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

$$c = \frac{1}{b-a}$$

b) $X \sim \text{exponential}(\lambda)$

$$f_X(x) = \lambda e^{-\lambda x} \quad x \geq 0$$

c) $X \sim \text{Normal}(\mu, \sigma)$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

→ (Cumulative Distribution Function).
Joint CDF of two R.V.s X, Y .

$$F_{X,Y}(x,y) = P(\{X \leq x\} \cap \{Y \leq y\})$$

we know that $\{X \leq x\} \in \mathcal{F}$.
 $\{Y \leq y\} \in \mathcal{F}$.

Properties of joint CDF

$$\textcircled{1} \quad \lim_{x \rightarrow \infty} F_{X,Y}(x,y) = F_Y(y).$$

$$\lim_{y \rightarrow \infty} F_{X,Y}(x,y) = F_X(x).$$

$$\lim_{x \rightarrow \infty} \lim_{y \rightarrow \infty} F_{X,Y}(x,y) = 1.$$

For discrete X, Y

$$F_{X,Y}(x,y) = \sum_{\substack{\hat{x} \leq x \\ \hat{y} \leq y}} P_{X,Y}(\hat{x}, \hat{y})$$

$$\lim_{n \rightarrow \infty} F_{X,Y}(x, y) = \sum_{\hat{y} \leq y} \left[\sum_{\hat{x} \in \mathcal{X}_0} P_{X,Y}(x, \hat{y}) \right]$$

$$= \sum_{\hat{y} \leq y} P_Y(\hat{y})$$

$$= F_Y(y).$$

$$\textcircled{2} \quad \lim_{n \rightarrow -\infty} F_X(x, y) = 0$$

$$\lim_{y \rightarrow -\infty} F_X(x, y) = 0.$$

$$\textcircled{3} \quad \text{if } x_1 \leq x_2, \quad y_1 \leq y_2$$

$$F_{X,Y}(x_1, y_1) \leq F_{X,Y}(x_2, y_2)$$

monotonicity property.

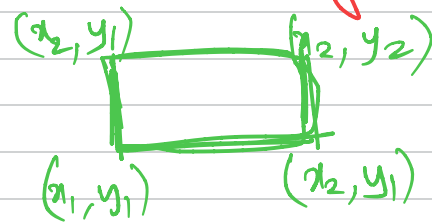
if $\omega \in A$
 $\Rightarrow X(\omega) \leq x_1, Y(\omega) \leq y_1$
 $\Rightarrow X(\omega) \leq x_2, Y(\omega) \leq y_2$
 $\Rightarrow \omega \in B$

$$A = \{X \leq x_1, Y \leq y_1\} \subseteq B = \{X \leq x_2, Y \leq y_2\}.$$

$$\textcircled{4} \quad \lim_{\substack{\epsilon \rightarrow 0^+ \\ \delta \rightarrow 0^+}} F_{X,Y}(x+\epsilon, y+\delta) = F_{X,Y}(x, y)$$

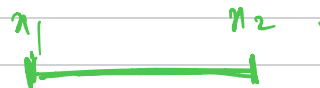
right continuity.

$$\textcircled{5} \quad P(x_1 < X \leq x_2, y_1 < Y \leq y_2)$$



$$= F_{X,Y}(x_2, y_2) - F_{X,Y}(x_2, y_1) - F_{X,Y}(x_1, y_2) + F_{X,Y}(x_1, y_1)$$

$$P(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$$



Continuity property

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{n \rightarrow \infty} P\left(\bigcup_{i=1}^n A_i\right)$$

Corollaries to this!

② if $A_1 \subset A_2 \subset A_3 \subset \dots$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{n \rightarrow \infty} P(A_n)$$

⑥ if $A_1 \supset A_2 \supset A_3 \dots$

$$P\left(\bigcap_{i=1}^{\infty} A_i\right) = \lim_{n \rightarrow \infty} P(A_n)$$

To prove $\lim_{n \rightarrow \infty} F_{X,Y}(x,y) = F_Y(y)$,

$$\overset{''}{\lim_{n \rightarrow \infty}} \boxed{F_{X,Y}(n,y)} = P(A_n) \text{ where}$$

$$A_n = \{\omega : X(\omega) \leq n, Y(\omega) \leq y\}$$

$$= \{X \leq n\} \cap \{Y \leq y\}$$

$A_1 \subseteq A_2 \subseteq \dots$ (increasing sets).

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} P(A_n) \therefore$$

$$= \lim_{n \rightarrow \infty} F_{X, Y}(n, y).$$

$$\bigcup_{n=1}^{\infty} A_n = \{\omega; Y(\omega) \leq y\} = \{Y \leq y\}.$$

||

$$\bigcup_{n=1}^{\infty} \{X \leq n\} \cap \{Y \leq y\}$$

$$= \{Y \leq y\} \cap \left[\bigcup_{n=1}^{\infty} \{X \leq n\} \right].$$

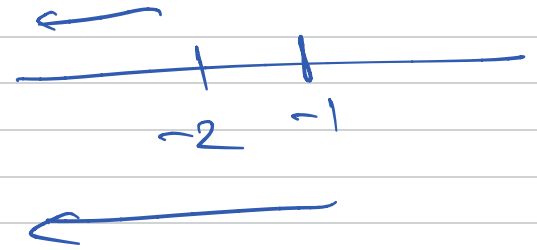
$$= \{Y \leq y\} \cap \Omega = \{Y \leq y\}$$

$$\Rightarrow P\left(\bigcup_{n=1}^{\infty} A_n\right) = P(\{Y \leq y\}) = F_Y(y).$$

②

$$\lim_{x \rightarrow -\infty} F_{X,Y}(x,y) = 0.$$

$$\stackrel{||}{\lim_{n \rightarrow \infty}} \boxed{F_{X,Y}(-n,y)} = P(A_n)$$



$$A_n = \{X \leq -n\} \cap \{Y \leq y\}.$$

$$A_1 \quad A_2 \quad A_3 \quad \dots \quad (\text{decreasing sets}),$$

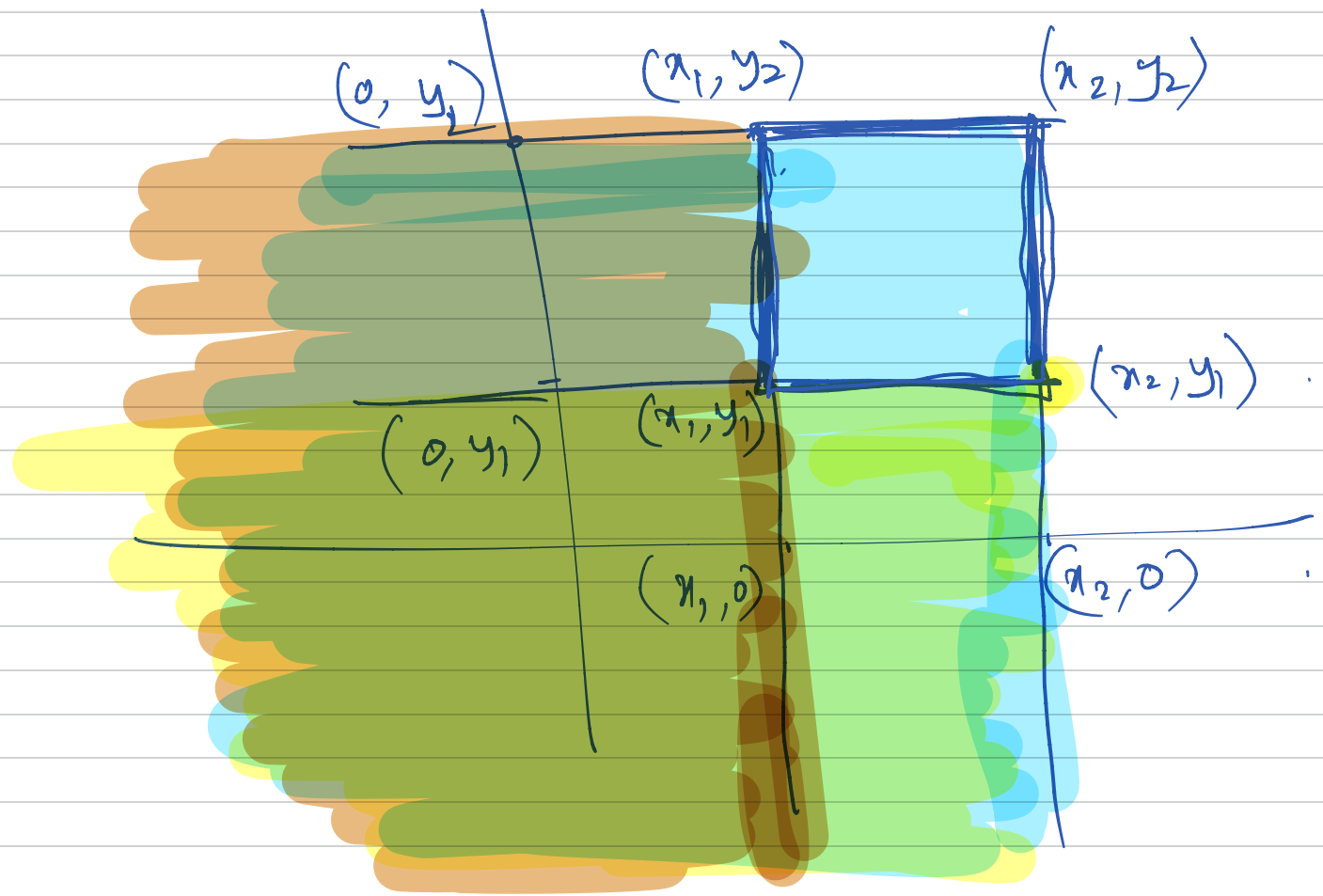
$$\lim_{n \rightarrow \infty} P(A_n) = P\left(\bigcap_{n=1}^{\infty} A_n\right)$$

$$\begin{aligned} \bigcap_{n=1}^{\infty} A_n &= \left(\bigcap_{n=1}^{\infty} \{X \leq -n\} \right) \cap \{Y \leq y\} \\ &= \emptyset \cap \{Y \leq y\} = \emptyset. \end{aligned}$$

$$\begin{aligned}
 \lim_{x \rightarrow -\infty} F_{X,Y}(x,y) &= \lim_{n \rightarrow \infty} F_{X,Y}(-n,y) = \lim_{n \rightarrow \infty} P(A_n) \\
 &= P\left(\bigcap_{n=1}^{\infty} A_n\right) \\
 &= P(\emptyset) = 0.
 \end{aligned}$$

⑤

$$\begin{aligned}
 &F_{X,Y}(x_2, y_2) \\
 &- F_{X,Y}(x_2, y_1) \\
 &- F_{X,Y}(x_1, y_2) \\
 &+ F_{X,Y}(x_1, y_1)
 \end{aligned}$$



Joint PDF of continuous R.Vs X, Y .

Joint PDF of continuous R.Vs X, Y is a function $f_{X,Y}(\cdot, \cdot)$ such that the Joint CDF

$$F_{X,Y}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u, v) du dv.$$

Suppose $F_{X,Y}(x, y)$ is differentiable in x, y then

$$f_{X,Y}(x, y) = \frac{\partial^2 F_{X,Y}(x, y)}{\partial x \partial y}.$$

Marginal pdf from joint PDF.

$$\int_{-\infty}^{\infty} f_{X,Y}(x,y) dx = f_Y(y)$$

$$\int_{-\infty}^{\infty} f_{X,Y}(x,y) dy = f_X(x)$$

From property 1 of Joint CDF.

$$F_Y(y) = \lim_{x \rightarrow -\infty} F_{X,Y}(x,y)$$

$$= \lim_{x \rightarrow -\infty} \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u,v) du dv$$

Discrete R.V.s

Joint PMF.

$$P_{X,Y}(x,y) = P(X=x, Y=y)$$

Marginal PMF

$$P_X(x) = \sum_y P_{X,Y}(x,y)$$

$$P_Y(y) = \sum_x P_{X,Y}(x,y)$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^y f_{X,Y}(u,v) du dv$$

$$= \int_{-\infty}^y \left[\underbrace{\int_{-\infty}^{\infty} f_{X,Y}(u,v) du}_{\text{pdf of } Y \text{ at } v.} \right] dv$$

$f_Y(v)$.

$$\Rightarrow f_Y(v) = \int_{-\infty}^{\infty} f_{X,Y}(u,v) du.$$

$$\boxed{f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx.}$$

Joint PDF properties

$$\textcircled{1} \quad f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy.$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx.$$

$$\textcircled{2} \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy$$

$$= \int_{-\infty}^{\infty} f_Y(y) dy = 1.$$

$$\textcircled{3} \quad f_{X,Y}(x, y) \geq 0 \quad \forall x, y \in \mathbb{R}.$$

Probability of rectangular region from joint PDF

$$P(x_1 \leq X \leq x_2, y_1 \leq Y \leq y_2) = \int_{x_1}^{x_2} \int_{y_1}^{y_2} f_{X,Y}(u,v) du dv$$

$$= F_{X,Y}(x_2, y_2) - F_{X,Y}(x_2, y_1) - F_{X,Y}(x_1, y_2) + F_{X,Y}(x_1, y_1)$$

$$= \int_{-\infty}^{x_2} \int_{-\infty}^{y_2} f_{X,Y}(x,y) dx dy - \int_{-\infty}^{x_2} \int_{-\infty}^{y_1} f_{X,Y}(x,y) dx dy$$

$$\int_{-\infty}^{x_2} \int_{y_1}^{y_2} f_{X,Y}(x,y) dx dy$$

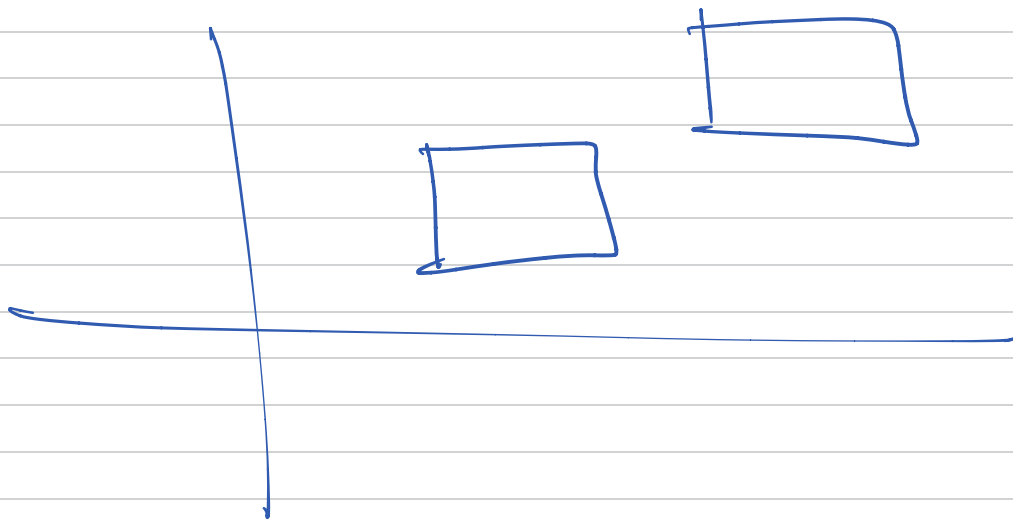
$$- \int_{-\infty}^{x_1} \int_{y_1}^{y_2} f_{X,Y}(x,y) dx dy \leftarrow$$

$$- \left[\int_{-\infty}^{x_1} \int_{-\infty}^{y_2} f_{X,Y}(x,y) dx dy - \int_{-\infty}^{x_1} \int_{-\infty}^{y_1} f_{X,Y}(x,y) dx dy \right]$$

$$= \int_{y_1}^{y_2} \int_{x_1}^{x_2} f_{X,Y}(x,y) dx dy.$$

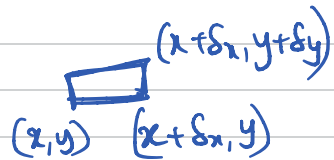
$$P((X,Y) \in B) = \int_{(x,y) \in B} f_{X,Y}(x,y) dx dy.$$

$\nwarrow$
 region



$$P(x \leq X \leq x + \delta x, y \leq Y \leq y + \delta y)$$

$$= \int_y^{y+\delta y} \int_x^{x+\delta x} f_{X,Y}(u,v) du dv.$$



$$= f_{X,Y}(x,y) \delta y \delta x.$$

$$\Rightarrow f_{X,Y}(x,y) = \frac{P((X,Y) \in [x, x+\delta x] \times [y, y+\delta y])}{\delta x \delta y}.$$

Joint density,
 $f_{X,Y}(x,y)$ is probability per area
 in the proximity of (x,y) .

Examples.

①. 2D Uniform R.V. $([0, 1] \times [0, 1])$

$$f_{X,Y}(x,y) = \begin{cases} c & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

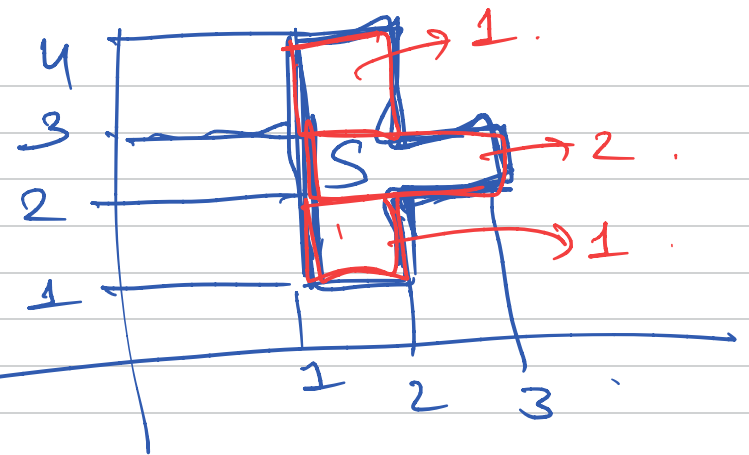
Find c .

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx dy = 1.$$

$$= \int_0^1 \int_0^1 c dx dy = 1$$

$$\Rightarrow c = 1.$$

② Uniform in region

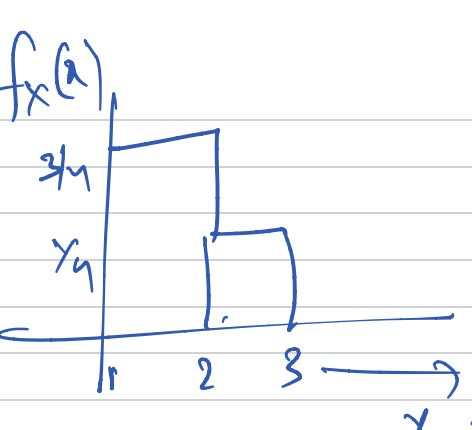


$$f_{X,Y}(x,y) = \begin{cases} c & \text{if } (x,y) \in S \\ 0 & \text{otherwise} \end{cases}$$

$$c \times 4 = 1 \Rightarrow c = 1/4.$$

Exercise: Find marginals of X, Y i.e., find $f_X(x)$ and $f_Y(y)$ for the above example.

$$\cancel{f_X}(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy.$$



$$f_X(x) = \begin{cases} \frac{3}{4} & x \in [1, 2] \\ \frac{1}{4} & x \in [2, 3] \\ 0 & x \notin [1, 3] \end{cases}$$

$$\int_{-\infty}^{\infty} f_X(x) dx = 1 = \int_1^2 \frac{3}{4} dx + \int_2^3 \frac{1}{4} dx$$

Exercise: Find $f_Y(y)$.